

## SAFETY DATA SHEET

Creation Date 03-June-2010

Revision Date 29-March-2024

Revision Number 3

### 1. Identification

**Product Name** (S)-(-)-Nicotine

**Cat No. :** L09879

**CAS-No** 54-11-5

**Synonyms** B-Pyridyl-a-N-Methylpyrrolidine.; S-(-)-Nicotine

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                           |            |
|---------------------------|------------|
| Acute oral toxicity       | Category 2 |
| Acute dermal toxicity     | Category 2 |
| Acute Inhalation Toxicity | Category 2 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Fatal if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray

Do not get in eyes, on skin, or on clothing

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

Wear respiratory protection

**Response**

IF ON SKIN: Wash with plenty of soap and water

IF INHALED: Remove person to fresh air and keep comfortable for breathing

Immediately call a POISON CENTER/doctor

Rinse mouth

Take off immediately all contaminated clothing and wash it before reuse

**Storage**

Store locked up

Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component | CAS-No  | Weight % |
|-----------|---------|----------|
| Nicotine  | 54-11-5 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms/effects**

No information available.

**Notes to Physician**

Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** 101 °C / 213.8 °F

**Method -** No information available

**Autoignition Temperature** 240 °C / 464 °F

### Explosion Limits

**Upper** 4.00 vol %

**Lower** 0.70 vol %

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Vapors may form explosive mixture with air. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
4

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep under nitrogen. Keep away from open flames, hot surfaces and sources of ignition. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component | Alberta                            | British Columbia                   | Ontario TWA/EV                     | Quebec                             | ACGIH TLV                          | OSHA PEL                                                                      | NIOSH                                                   |
|-----------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------|
| Nicotine  | TWA: 0.5 mg/m <sup>3</sup><br>Skin | TWA: 0.5 mg/m <sup>3</sup><br>Skin | TWA: 0.5 mg/m <sup>3</sup><br>Skin | TWA: 0.5 mg/m <sup>3</sup><br>Skin | TWA: 0.5 mg/m <sup>3</sup><br>Skin | (Vacated) TWA:<br>0.5 mg/m <sup>3</sup><br>Skin<br>TWA: 0.5 mg/m <sup>3</sup> | IDLH: 5 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments                 |
|----------------|-------------------|-----------------|--------------------------------|
| Nitrile rubber | < 108 minutes     | > 0.12 mm       | As tested under EN374-3        |
| Neoprene       |                   |                 | Determination of Resistance to |
| Natural rubber |                   |                 | Permeation by Chemicals        |
| PVC            |                   |                 |                                |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties**

|                                         |                              |
|-----------------------------------------|------------------------------|
| <b>Physical State</b>                   | Liquid                       |
| <b>Appearance</b>                       | Colorless, Yellow, Brown     |
| <b>Odor</b>                             | Fishy                        |
| <b>Odor Threshold</b>                   | No information available     |
| <b>pH</b>                               | 10.2 0.05M aq.sol            |
| <b>Melting Point/Range</b>              | -80 °C / -112 °F             |
| <b>Boiling Point/Range</b>              | 247 °C / 476.6 °F @ 760 mmHg |
| <b>Flash Point</b>                      | 101 °C / 213.8 °F            |
| <b>Evaporation Rate</b>                 | No information available     |
| <b>Flammability (solid,gas)</b>         | Not applicable               |
| <b>Flammability or explosive limits</b> |                              |
| <b>Upper</b>                            | 4.00 vol %                   |
| <b>Lower</b>                            | 0.70 vol %                   |
| <b>Vapor Pressure</b>                   | 0.53 mbar @ 25 °C            |
| <b>Vapor Density</b>                    | 5.61                         |

|                                        |                          |
|----------------------------------------|--------------------------|
| Specific Gravity                       | 1.015                    |
| Solubility                             | miscible                 |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | 240 °C / 464 °F          |
| Decomposition Temperature              | 200°C                    |
| Viscosity                              | No information available |
| Molecular Formula                      | C10 H14 N2               |
| Molecular Weight                       | 162.23                   |

## 10. Stability and reactivity

|                                  |                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                  |
| Stability                        | Stable under normal conditions.                                                             |
| Conditions to Avoid              | Incompatible products. Excess heat.                                                         |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Strong bases                                         |
| Hazardous Decomposition Products | Nitrogen oxides (NO <sub>x</sub> ), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                    |
| Hazardous Reactions              | None under normal processing.                                                               |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component | LD50 Oral                                           | LD50 Dermal                                          | LC50 Inhalation            |
|-----------|-----------------------------------------------------|------------------------------------------------------|----------------------------|
| Nicotine  | ECHA (RAC) ATE = 5 mg/kg<br>LD50 = 50 mg/kg ( Rat ) | ECHA (RAC) ATE = 70 mg/kg<br>LD50 = 50 mg/kg ( Rat ) | ECHA (RAC) ATE = 0.19 mg/l |

Toxicologically Synergistic Products No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | May cause skin, eye, and respiratory tract irritation                                    |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|---------|------------|------------|------------|------------|------------|
| Nicotine  | 54-11-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects Substances known to cause developmental toxicity in humans.

Teratogenicity No information available.

STOT - single exposure None known

STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects, both acute and No information available

delayed

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Algae | Freshwater Fish                            | Microtox   | Water Flea                             |
|-----------|------------------|--------------------------------------------|------------|----------------------------------------|
| Nicotine  | Not listed       | LC50: 4 mg/L/96h<br>(Onchorhynchus mykiss) | Not listed | EC50: 0.24 mg/L/48h<br>(Daphnia pulex) |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component | log Pow |
|-----------|---------|
| Nicotine  | 1.17    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component          | RCRA - U Series Wastes | RCRA - P Series Wastes                                                                                             |
|--------------------|------------------------|--------------------------------------------------------------------------------------------------------------------|
| Nicotine - 54-11-5 | -                      | does not include patches, gums, and lozenges that are FDA-approved over-the-counter nicotine replacement therapies |

## 14. Transport information

### DOT

UN-No UN1654  
 Proper Shipping Name NICOTINE  
 Hazard Class 6.1  
 Packing Group II

### TDG

UN-No UN1654  
 Proper Shipping Name NICOTINE  
 Hazard Class 6.1  
 Packing Group II

### IATA

UN-No UN1654  
 Proper Shipping Name NICOTINE  
 Hazard Class 6.1  
 Packing Group II

### IMDG/IMO

UN-No UN1654  
 Proper Shipping Name NICOTINE  
 Hazard Class 6.1  
 Packing Group II

## 15. Regulatory information

**International Inventories**

| Component | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Nicotine  | 54-11-5 | X   | -    | X    | ACTIVE                                        | 200-193-3 | -      | -   |

| Component | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|---------|-------|----------|------|------|------|------|-------|-------|
| Nicotine  | 54-11-5 | X     | KE-25325 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nicotine  | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

**REACH links**<https://echa.europa.eu/substances-restricted-under-reach>**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component | CAS-No  | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|---------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Nicotine  | 54-11-5 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Nicotine  | 54-11-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Product Safety Department  
 Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)  
[www.thermofisher.com](http://www.thermofisher.com)

**Creation Date**

03-June-2010

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Revision Date</b>    | 29-March-2024                                      |
| <b>Print Date</b>       | 29-March-2024                                      |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

**Disclaimer**

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**End of SDS**